1. Trading

As a solution to this exercise, the code TechnicalTest_Trader_MiguelManrique.ipynb which has the analysis is attached to this document.

Handling 0.0 Values

While we could apply various data cleaning techniques such as interpolation, averaging, or the last observation carried forward, I have decided to omit them. This is due to simplicity and the fact that these methods can alter data integrity and reliability, potentially leading to erroneous hypotheses—especially in time-series data where, under certain conditions and over short periods, there are no significant variations. Furthermore, addressing this would require a separate, comprehensive study that tends to be exhaustive.

Handling Outliers

By sorting the data by index and plotting it, we can observe a period prior to September 2020 with erratic data, which is unusual for financial time series. Given that I lack access to external information for further context (e.g., takeovers, splits, etc.), it is difficult to propose a data normalization that does not compromise their integrity, potentially leading to skewed distributions and erroneous hypotheses. Consequently, I opted to preserve only the data from April 2017 onwards.

Price Time Series Analysis

By using the typical price to filter out the noise generated by intraday highs and lows, we can identify a significant **support level around 19k**. With a sustained bullish trend since mid-2023, the price is currently approaching a **resistance zone above 60k**. Given that the **RSI is above the 70 mark (overbought)**, caution is advised for long positions; it is recommended to wait for a retracement as a potential buying opportunity.

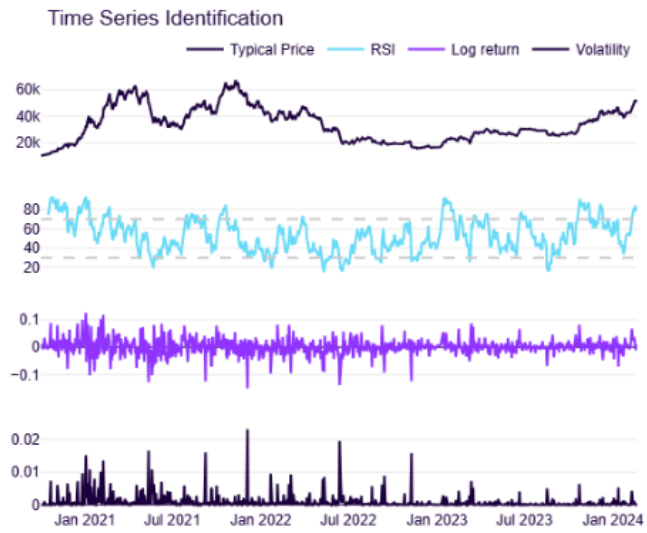

Chart 1.1 Typical Price, RSI, Log Return and Volatility

Trading Strategies and Back testing

As an exploratory exercise, two data-driven trading strategies were proposed. The first is a simple Bollinger Bands strategy, aiming to buy when the price is below the lower band and sell when it reaches the upper band, maintaining strict position sizing control not exceeding one unit.

Chart 1.2 Bollinger Bands Trading Strategy, moving and stadard deviation of 20 periods and 2 deviations from ma.

The second strategy consists of a Support Vector Machine (SVM) model that aims to estimate the price's next direction. Thus, an estimate of 1 indicates an expected upward movement, while -1 indicates a downward movement; this strategy also maintains strict risk control limited to a single position.

Chart 1.3 Support Vector Machine Trading Strategy, using 8 lags, 30 as a regularization parameter and rgb kernel.

Back-test

As an exploratory exercise, this backtest can be considered a starting point for refining each strategy. For instance, by using cross-validation, we can optimize parameters such as the lag periods for Bollinger Bands or the regularization coefficient (C) and kernel type for the SVM. Furthermore, risk constraints could be incorporated, such as a maximum stop-loss percentage per position or limiting the trade frequency to optimize transaction costs.

	Bollinger Bands	SVM
Total Return	-16,50%	30,21%
Sharpe Ratio	-47,05%	18,38%
Max Drawdown	-18,13%	-53,72%

Table1.1 Back-test

Block-chain

Blockchain is based on a chain of encrypted, immutable blocks, distributed across multiple 'nodes' (personal PCs, servers, and data centers). Its core is data cryptography, and it is comprised of various types of agents throughout the entire network.

Is the foundation of the Web3 revolution, enabling the decentralization of human activities related to the exchange of goods and services, as well as the protection of intellectual property such as patents, artworks, and gaming assets. In the financial sector, this allows for reduced costs and faster transaction times for monetary units and, in the future, for tokenized assets, while maintaining the confidentiality and security of the agents involved. For instance, the article (Muyao Shen, 2026)promotes the integration of blockchain with one of the world's largest stock exchanges

2.1 USDCOP FX Risk, execution and hedge design

The solution of this exercise assumes a **constant 10% FX tolerance band** (±5% upside/downside risk) and **70%** hedge ratio as core compliance parameters, favoring a tighter band and an increased hedge ratio.

Given that the NDF curve **trading at a premium**, the strategy seeks to optimize long USD positions while minimizing the **cost of carry** for short positions. A layered strategy is implemented by splitting the hedge notional into four weekly 25% tranches. For week 8, since the hedge requires an 11-week NDF, we use a custom NDF curve derived from SOFR and IBR rates adjusted to their respective maturities.

Week	SOFR	IBR	Spot	Indicative NDF
9	3,78%	12,00%	4.058,0	**4115,4**
10	3,78%	12,00%	4.058,0	**4121,7**
11	3,78%	12,00%	4.058,0	**4128,0**
12	3,78%	12,00%	4.058,0	**4134,3**

Table 2.1.1: Estimación NDF curve semana 9 a 12

Exposure Map

From week 1 to 8, the average gross hedge amount was **$539k**, peaking at **$1.06M** in week **8**. Weekly reserve income and hedge cash margins averaged **$28.7k** and **$228k**, respectively. The USD mint/redeem surplus averaged **$475k** (ranging from $1.9M to $2.6M), while the COP surplus averaged COP **$437.5M** with a minimum of $2.6B and maximum of $2.8B

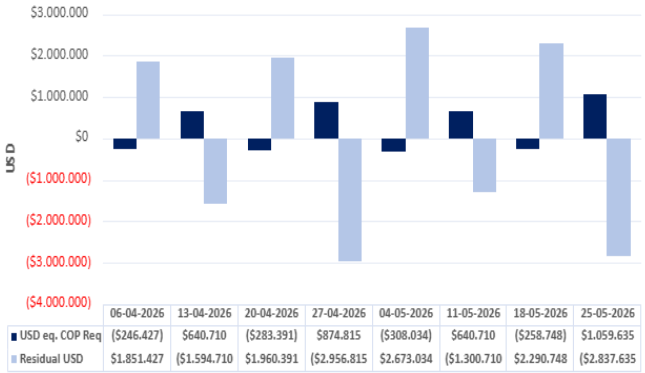

Chart 2.1.1: Exposure map, weekly mandatory COP needs and residual USD liquidity after meeting COP needs.

week 1 to 8	Mean	Std	Min	Max
USD eq. COP req	$539.059	$313.356	$246.427	$1.059.635
USD reserv inc	$28.750	$2.659	$25.000	$32.000
USD Hedge Cash	$228.125	$40.527	$180.000	$300.000
USD surplus	$475.000	$1.857.610	($1.900.000)	$2.600.000
COP surplus *	$437,5	$2.395	($2.600)	$2.800

*Table 2.1.2: Treasury performance week 1 to 8, * COP millions*

Comparative Analysis: Fully Static, Layered and Rule-Based Dynamic Hedging

The **layered hedging** strategy distributes 25% of the required coverage notional across the next four NDFs on the curve, incorporating estimates for weeks 9, 10, and 11, which were not in the initial dataset. It is propose a **rule-based dynamic hedging** approach that combines fully static and layered based on portfolio's directional bias:

- **Short USD Position:** A fully static hedge at the corresponding fixing to lock in the current rate and eliminate delta risk.
- **Long USD Position:** A layered hedge by averaging the NDF rates across the next four settlements.

The comparison of these three methodologies under base, bull, and stress scenarios is summarized in the table below.

Hedge P&L	Base Fixing	Bull COP	Stress COP
Fully Static Hedge	24.919	-6.383	64.780
Layered Hedge	26.775	-4.468	66.586
Ruled - based Dynamic Hedge	27.283	-3.978	67.082

Table 2.1.3: Hedging programs comparative analysis under 3 case scenarios.

It is concluded that the rule-based dynamic hedge, leveraged by the NDF curve structure, **optimizes coverage** efficiency across all market scenarios.

Furthermore, a **simplified VaR** (Value at Risk) analysis, based on historical base fixing returns shows an average return of **0.43%,** which remains well within the 10% FX tolerance band. This supports our recommendation to integrate this finding into a dynamic hedging rule: if the next NDF tenor exceeds this historical variation, the hedge ratio should be increased proportionally to the NDF's deviation from the average base fixing return.

Finally, based on the provided data, there is a 95% confidence level that weekly treasury losses **will not exceed** 2.5% relative to the **hedge cash margin.**

	Non-Hedge	Hedge (70%)
VaR (95%)	$8.120	$5.684
VaR / Hedge Cash	3,56%	2,49%
VaR / Reserve Income	28,24%	19,77%

Table 2.1.4: VaR estimation.

Hedge Carry and P&L under Basis and Stress COP scenarios in ruled-based dynamic program

The largest one-week liquidity requirement was on week 8, at USD $1.059M.

Week	Cost Of Carry Base	Cost of Carry Stress	P&L Base	P&L Stress
1	-4	-117	$1.186	($3.504)
2	-17	-129	$3.174	$15.021
3	10	-95	$2.052	($2.971)
4	-24	-142	$5.335	$22.189
5	12	-108	$2.295	($3.887)
6	-32	-149	$4.735	$16.851
7	8	-122	$372	($5.153)
8	-34	-154	$8.134	$28.535
Total			**$27.283**	**$67.082**

Table 2.1.5: Base and Stress scenarios comparison in ruled-based dynamic hedge program

Based on the daily controls below, can determine which **NDF** best suits our hedging strategy. The **ticket size (volume)** should approximate the Average Daily Volume (ADV), identifying liquidity accumulation levels through technical analysis (support and resistance, book depth, liquidity structure, and order flow). **Tolerance bands** are governed by risk limits, such as Daily VaR (Value at Risk)

	Position
USD	$1.000.000
COP (M)	$3.500
TRM	$4.020
Exposure (USD to DO)	**$870.647**

Limits	Value	Threshold
Max exposure	$2.350.000	$10.000.000
Hedge ratio	85%	70%
VaR (99%) week	1,34%	2,50%

Table 2.1.6 Position and exposure monitor

NDF	Indicative
Week 1	4068,00
Week 2	4081,00
Week 3	4095,00
Week 4+	4108,00

Rates	Value
OIS	9,87%
IBR	10,20%
TIB	9,60%
DTF	8,60%
SOFR	4%
Tbills	4,10%

Table 2.1.7 NDF curve and Interest rates monitor

Market	USDCOP
BID	3702,00
ASK	3706,00
Spread	0,11%
Max	3725,00
Min	7677,00
VWAP	3704,87
Daily Volume (M)	$1.361
Avg Daily Volume	$743.780

Table 2.1.8 FX market monitor

Furthermore, my daily control must include **technical analysis** and fundamentals such as Banrep and Hacienda movements, global trends and internal flows

Chart 2.1.2 Price chart

2.2 Reserve Trust design and investment manual

The solution for this exercise proposes the implementation of a **Linear Programming (LP) model,** drawing a parallel to the Markowitz Portfolio Optimization framework. The model defines de objective funcion aimed at maximizing the **weighted average yield** subject to a multi-layered constrain matrix, ensuring full compliance with liquidity, haircut coverage, internal concentration cap, and weighted average life constrains (as detailed in Appendix 2.2)

Policy clauses and practical reserve design

According to global treasury standards[1], liquidity management must balance **capital preservation**, **risk mitigation**, and **yield optimization**, among others company business objectives. (JP Morgan, 2026) suggests that a robust treasury framework should segment cash flows into three distinct liquidity needs:

- **Operating:** Required for same-day liquidty and immediate obligations
- **Reserve:** Assets held for 6 to 12 liquidity horizons
- **Strategic:** Long-term investments with a 12+ month horizon

[1] (Dashmeet Kaur, 2024; JP Morgan, 2026)

Aligned with our current case, the **Investment Policy Statement (IPS)** for the operating cash flow is governed by the following pillars:

Liquidity Management ensures an appropriate liquidity control to fully cover the mint/redemption COPW needs monitored through the liquidity risk index (**IRL**)[2].

Reserve Trust According to the principles of a good reserve management in IMF y Basilea III[3] a strategy of cash allocation may consider:

1. **High Quality Liquid Assets (HQLA) principle.** Investments are restricted to high-grade instruments such as AAA, AA+, AA- (bonds) and Tier-1 counterparty bank, AAA MMF, Sovereign, primary dealer, etc.
2. **Immediate liquidity principle.** A calculated reserve percentage that satisfies estimated short-term obligations estimated by a dynamic draw up cash forecast.
3. **Segregation & Concentration principle.**

   Assets are legally and operationally segregated. To Ensure immediate availability, sight cash accounts (0-day WAL) are prioritized. Furthermore, a 50% concentration cap per instrument and counterparty is enforced to mitigate idiosyncratic and counterparty risk.

Under this framework the methodology of universe selection is summarized as the chart below

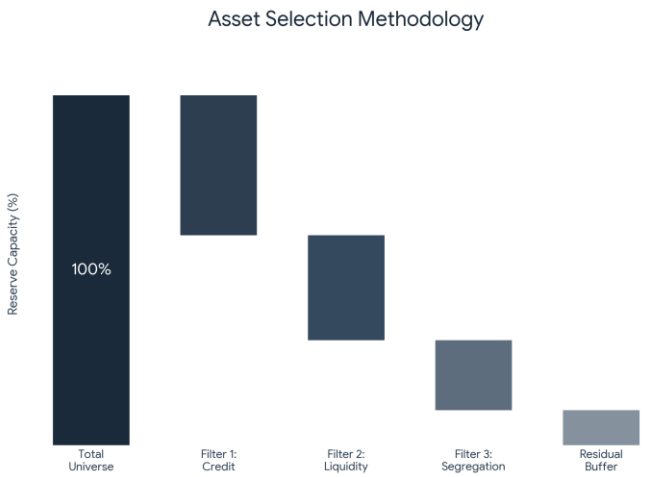

Figure 2.2.1: Hierarchical screening process for reserve assets based on credit quality, immediate liquidity, and legal segregation thresholds.

It could finally get set up the universe as

Asset / instrument	HQLA	Liquidity	Seggragation
Local bank sight cash	x	x	x
COP money market fund	x	x	x
TES 0-90d	x	-	x
Collateralized COP repo	x	x	x
COP term deposit 30d	x	-	x
Unsecured finance co. deposit			x

Table 2.2.1 Asset/Counterparty selection criterion

In adherence to the **(IPS), Unsecured Finance Co. Deposits** are excluded from the eligible universe to satisfy the HQLA principle, **30-day COP Term Deposits** and **0-90 day TES** are subject to concentration limits under Immediate Liquidity Principle, ensuring the do not compromise the same-day availability required for COPW redemptions.

To mitigate **breach escalation risk,** a monitor over key risk indicators including credit ratings, weighted average life (WAL) and portfolio allocation. Any deviation triggers a protocol to the Treasury Manager and CFO for remediation. In the event of a **redemption waterfall scenario** the liquidation hierarchy is priorized as follows: 1) Local bank sight cash accounts, 2) COP Money Market Maker Funds, 3) Collateralized COP Repos.

Applying this IPS and reserve design the target reserve mix is propose as follows

Target Reserve Mix

Asset / instrument	Eff. COP wt.
Local bank sight cash	30,0%
COP money market fund	25,0%
TES 0-90d	0,0%
Collateralized COP repo	25,0%
COP term deposit 30d	20,0%
Unsecured finance co. deposit	0,0%

Table 2.2.2 Asset weights

Compliance Checker

Metric	Value	Threshold	Status
COP weights = 100%	100,0%	100,0%	OK
COP weighted yield	9,925%		
COP same-day liq.	97,0%	40,0%	OK
COP weighted WAL	7	60	OK
COP h/c adj. cov.	99,575%	**99,0%**	OK

Table 2.2.3 Asset allocation metrics

Haircut adjusted coverage modification explanation

A COP minimum haircut-adjusted coverage has been implemented, setting the threshold at 99%. Based on the provided dataset, achieving a 101% adjusted coverage is mathematically infeasible under current constraints.

However, in strict adherence to our **Reserve Design and Policy Clauses**, this modification is risk-justified by our liquidity position: with a COP minimum same-day liquidity threshold of 40%, the reserve maintains enough cash to cover x3.2 times the estimated daily liquidity needs. Consequently, the reduction from 101% to 99% in haircut does not compromise the trust's solvency, as the surplus in cash provides robust buffer against short-term redemption volatility.

Reporting Pack

As an expansion of Table 2.2.3, the treasury monitoring suit should be augmented by incorporating **Credit Rating, Weigthed Average Life (WAL), and Portfolio Allowance**

[2](SFC, 2018)

[3] (Comité de Supervisión Bancaria de Basilea, 2010; International Monetary Fund, 2014)

metrics per instrument. Furthermore, integrating an IRL Tracker, COPW mint/redeem logs, and dynamic cash flow forecasting will improve liquidity oversight.

2.3 Cross-chain liquidity management and on-chain arbitrage

To fulfill the objectives of this exercise, the following are presented: **i) an optimization method for the constant product function** utilized by liquidity pool to calculate the potential slippage; this serves as the core business logic for **ii) a practical arbitrage algorithm in Python** (the source code is provided in the attached .zip file).

Rebalancing Plan

An optimal rebalancing plan must accomplish the boundaries by transferring the minimum amount of funds.

Chain	Net COPW transfer (+out / -in)	Net tCOPW transfer (+out / -in)
Ethereum	$-300.000	$-300.000
Base	0	0
Polygon	$-200.000	$200.000
Solana	0	0

Table 2.3.1 rebalancing transactions

On-chain arbitrage

The inventory rebalancing and arbitrage strategy is modeled as a **Quadratic Programming (QP) minimization problem** (see Appendix 2.3). Given a specific pool mid-price, the objective function aims to determine the optimal allocation of quote currency to maximize net P&L under transaction cost constrains.

Chain	tCOPW in (buy tCOPW)	Net P&L vs redeem	COPW in (sell COPW)	Net P&L vs mint
Ethereum	20.173	$189,91	-	-
Base	-	-	15.200	$80,25
Polygon	10.000	$52,12	-	-
Solana	-	-	22.559	$306,56

Table 2.3.1 arbitrage in-flows and out-flows

Algorithmic arbitrage

A recommended starting architecture for an algorithmic arbitrage bot should consider three key factors: 1) **Infrastructure** for high-frequency trading, C++ is preferred over python. The environment must be resilient and low-latency, leveraging direct Exchange API's and WebSocket feeds to ensure real-time synchronization, 2) **Strategic business logic,** 3) **End-to-End operation control** a comprehensive front-to-back office integration is important to reconcile the chain inventory maintaining full visibility over the reserve desk's exposure.

For this practical exercise is attached a low complexity algorithm which mimic chain market data; also, the following recommendation could be taken as an approximation framework to stablish triggers, sizing method, price bands, pre-tade checks, post-trade reconciliation and kill switches.

Triggers execution is based on a net P&L threshold, e.g. an order is triggered only if Spread > (AMM_fee + Slippage + Gas + Bridge_costs) - "sniper algo".

Sizing methods consider the 5-day average daily volume and current pool reserve to minimize price impact; also size is inversely proportional to expected slippage to stay within market risk tolerance.

Price bands Similar to a bid/ask spread in TradFi an On-chain peg variance band assures the algorithm only operates when the COPW price deviates beyond the cost of a mint/redeem cycle.

Pre-trade checks automated verification of inventory (available COPW/rCOPW vs target bands), bridge status (latency and uptime of the chain route), T-1 reconciliation ensuring no "ghost" positions from pending transaction

Post-trade reconciliation real-time drop copy of transaction hashes, use of webhook and back-office notification at day end.

Kill switches are design for catastrophic scenarios, providing a failsafe mechanism to decommission the algorithm systematically. This involves cancelling pending or working orders, recovery roadmap to regain operation control, ensuring the integrity of the system before a reboot.

Circuit breakers are implemented as automated safeguards to pause algorithmic execution when predefined risk or config thresholds are breached (e.g. daily maximum reserve capacity per chain, halted status in a chain, volatility, among others).

An effective way to prevent reserve desk mismatches is by implementing trade confirmation alerts via webhooks. When a trade is executed, an automated notification is triggered for the desk; this message contains all relevant transaction data, such as the hash ID, size, price, and order confirmation ID.

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Appendices

Appendix 2.2 Quantitative framework for reserve trust optimization

1. Linear Programming (LP) objective function

$$\max \quad R = \sum_{i=0}^{n} y_i w_i$$

Subject to

$$\sum_{i=0}^{n} w_i = 1,$$

$$\sum_{i=0}^{n} l_i x_i \geq Liquidity\,Threshold$$

$$\sum_{i=0}^{n} hc_i x_i \geq Haircut\,Threshold$$

$$w_i \geq 0\ \forall_i$$

$$w_i \leq Internal\,Weight\,threshold$$

$$x_{wal,i} \leq WAL\,Threshold$$

$$\ldots$$

Appendix 2.3 Quantitative Framework for Liquidity Optimization

1. Quadratic Programming (QP) Objective Function

$$\min_{x} \quad -[Income(\Delta x) - \Delta x - Fees(\Delta x) - Gas]$$

Were

$$Out(\Delta x) = Y - \frac{X \cdot Y}{X + \Delta x \cdot \left(1 - fee_{lp}\right)}$$